# CREDIT RISK MANAGEMENT AND REGULATION

## (ESTIMATING DEFAULT PROBABILITIES, CREDIT VALUE AT RISK, COUNTERPARTY CREDIT RISK, REGULATORY REQUIREMENTS)

Sources:

- *Risk Management course – Lecture 7, 8 & 9 slides*
- *Risk Management and Financial Institutions,* Hull; Ch 19, 21
- *https://www.tnpconsultants.com/wp-content/uploads/2023/06/Merton-Model-in-Credit-Risk-Modelling-version-2023.pdf*

## Credit Risk

**Definition: Credit risk is the *risk of loss arising from the non-performance of counterparties***, i.e. the risk of loss resulting from *partial or total failure to meet obligations* (on or off-balance sheet) to the financial institution. Typically, it is the largest component of banking risk.

Main credit risk subtypes:

- **credit default risk**: the credit risk associated with certain financial or investment services activities (e.g. lending, borrowing, financial leasing)
- **counterparty credit risk**: the potential loss arising from the failure of a counterparty to a given (typically derivative) transaction to meet its contractual obligations before the transaction is closed (final settlement of cash flows).
- **issuer credit risk:** the credit risk to the issuers of debt securities and equity securities, which includes the risk that the value of the security will decline because of a deterioration in the issuer's credit quality.
- **country risk** etc.

**Credit risk concepts, Measuring Credit Risk:**

- **Default event**: the counterparty to a financial contract (e.g. loan contract, security (bond), derivative) fails to perform as contractually agreed
- Default risk **probability of default** (**PD**, %):
    - the *probability that a default event will occur for a given counterparty/contract within a given time period* (e.g. one year)
- **Loss Given Default** (**LGD**, %) = 1 - Recovery rate

- *Percentage of loss in case of default*

- **Exposure** (at Default, **EAD**): exposure, *the total amount of claims outstanding at the time of default*.
- **Expected (Credit) Loss (ECL or EL) = PD*LGD*EAD**

**PD Estimation**

The probability of default can be estimated using *historical data, credit spreads* or *equity models*.

***1)*** *Historical data (obtained from rating agencies or a bank's own data)*

Rating agencies provide ratings describing the creditworthiness of corporate bonds. They usually rate companies that issue publicly traded debt. Credit ratings are revised infrequently (bc bond traders are major users of credit ratings, they would frequently need to trade), only when there is a reason to think that long-term change in the company's creditworthiness has taken place. These are *external credit ratings*, however, for example, banks that do not have publicly issued debt use internal credit ratings, rating the creditworthiness of their corporate and retail clients. Banks can use their internal rating to calculate PD also for themselves (Basel II). Estimated PDs by this method are *real-word probabilities*.

Cumulative Average Default Rates are provided by rating agencies.

| | Time (years) | | | | | | |
|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | 7 | 10 |
| Aaa | 0.000 | 0.013 | 0.013 | 0.037 | 0.104 | 0.241 | 0.489 |
| Aa | 0.022 | 0.068 | 0.136 | 0.260 | 0.410 | 0.682 | 1.017 |
| A | 0.062 | 0.199 | 0.434 | 0.679 | 0.958 | 1.615 | 2.759 |
| Baa | 0.174 | 0.504 | 0.906 | 1.373 | 1.862 | 2.872 | 4.623 |
| Ba | 1.110 | 3.071 | 5.371 | 7.839 | 10.065 | 13.911 | 19.323 |
| B | 3.904 | 9.274 | 14.723 | 19.509 | 23.869 | 31.774 | 40.560 |
| Caa | 15.894 | 27.003 | 35.800 | 42.796 | 48.828 | 56.878 | 66.212 |

The table shows the probability of default for companies *starting with a particular credit rating*.

A company with an initial credit rating of Baa has a probability of 0.174% of defaulting by the end of the first year, 0.504% by the end of the second year, and so on.

- For investment-grade bonds, the PD in a given year tends to be an increasing function of time (greater chance that in some years will have financial problems). For bonds with poor credit rating, it is a decreasing function of time (the next 1-2 years is critical, if survives, good).
- The **unconditional default probability** is the probability of default during a given year as seen at time zero.

- The **hazard rate** or default intensity is the probability of default over a short period of time, conditional on no earlier default. ($V_t$ is the cumulative probability of the company surviving until $t$):

$$\frac{V(t) - V(t + \Delta t)}{V(t)} = \lambda(t)\Delta t$$

$$V(t) = e^{-\int_0^t \lambda(\tau)d\tau}$$

$$Q(t) = 1 - e^{-\bar{\lambda}(t)t}$$

  It is the unconditional probability of default during year $t$ over the probability of surviving until year $t$.

- **Recovery Rate**: The recovery rate for a bond is normally defined as *the price at which it trades 30 days (about one month) after default* as a percent of its face value; **Recovery rate = 1 – LGD**. The recovery rate is negatively related to default rates (in a year with low number of defaults the economic conditions are usually good, so the recovery rate is higher, and vice versa). We can estimate average recovery rates from historical data, *reported by rating agencies*.

***2)*** *Credit spreads (e.g., CDS spreads, bond yield spread)*

**Credit Default Swaps (CDS):**

CDS is a derivative that allows market participants to *trade credit risks*. It is an *instrument that provides insurance to the buyer against the risk of a default by a particular company.* This company is the *reference entity*, its default is known as a *credit event* (failure to make a payment as it becomes due, restructuring of debt, or bankruptcy). The buyer of the insurance obtains the right to sell bonds for their face value when a credit event occurs, and the seller of the insurance agrees to buy them. The total face value that can be sold is the CDS's *notional principal*. The buyer makes *periodic payments* to the seller (usually quarterly) called *premium* until the end of the life of the CDS or until a credit event. The total amount paid in a year, as a percent of the notional, is the *CDS spread*. In case of a credit event, accrued payment has to be made by the buyer before the seller pays. Contracts usually mature on predetermined, standard dates. If there is a default, the buyer has the right to sell bonds (*physical settlement*) or get compensated in cash for the loss (*cash settlement*).

CDS is different than other derivatives, bc as CDSs depend on default probabilities, and some market participants may have better information to estimate this probability than others – *asymmetric information*.

- **CDS spread:** CDS can be used to hedge a position in a corporate bond. The effect of a CDS is to *convert the corporate bond to a risk-free bond*: on a 5y corporate bond yielding 7% per year you can buy CDS protection for 200bps per annum. The n-year CDS spread should be approx. equal to the excess of the par yield on an n-year corporate bond over the par yield on an n-year risk-free bond.

**Using credit spread to Predict PD:**

Credit spreads can be used to predict PD. For this CDS spreads, bond yield spreads or asset swap spreads could also be used. Bond yield spreads could be used if the bond sells close to par value. The bonds yield spread should be close to the CDS spread, o.w. there would be arbitrage.

- **Average hazard rate** (avg PD per year over the total period, cond. on no early default) between time 0 and $t$ is a good approximation:

$$\bar{\lambda} = \frac{s(t)}{1-R}$$

where *s(t)* is the credit spread calculated for a maturity of $t$, and $R$ is the recovery rate (1-LGD).

- For a 1-year CDS: **PD(1yr) = CDS spread(1yr) / LGD**

***3)*** *Structured (equity) models (based on economic concepts behind default, e.g., Merton model)*

Equity prices provide more up-to-date information for estimating PD because credit ratings are revised infrequently, only when there is a reason to think that long-term change in the company's creditworthiness has taken place.

**Merton Model:**

The model uses *options pricing theory* to evaluate the likelihood that a firm's assets will fall below its liabilities at a future point in time, leading to default. In this model the company's shareholders have a *long call option on the company's assets* with a strike price equal to the repayment required on debt, D (zero bond), at maturity, T. If at maturity $V_T < D$, a default occurs and creditors take over the company (liquidation cost is zero), making this a *short put* option from their perspective, with a payoff:

$$\mathbf{D_T = \min(A_T, D_T)}$$

The value to shareholders in this case is zero. If $V_T > D$, the value of the equity after fulfilling debt payments is $V_T - D$. Under the model the company defaults when the call option is not exercised. The options' payoff function (value of shares at debt maturity) is:

$$\mathbf{E_T = \max(V_T - D, 0)}$$

This call option (stock price, share of leveraged company) can be priced using the *Black-Scholes formula (1)*:

$$E_0 = V_0 N(d_1) - De^{-rT} N(d_2)$$

where:

$$d_1 = \frac{\ln(V_0/D) + (r + \sigma_V^2/2)T}{\sigma_V \sqrt{T}}\ ;\ d_2 = d_1 - \sigma_V \sqrt{T}$$

The value of the firm $V_t$ follows a GBM:

$$\frac{dV_t}{V_t} = rdt + \sigma_V \, dW_t$$

$$dV_t = r V_t dt + \sigma_V V_t dW_t$$

***N(-d₂)*** *= 1 – N(d₂)* is the **probability of default** (option not exercised).

$$PD = N\left(\frac{\ln\left(\frac{D_T}{V_T}\right) - \left(r - \frac{\sigma_V^2}{2}\right)/T}{\sigma_V \sqrt{T}}\right)$$

To calculate this we need $V_0$ and $\sigma_V$ but none of these are observable. If the company is publicly traded, we can observe $E_0$. Using this, we can estimate $\sigma_E$ through *Ito's Lemma (2)*:

$$\sigma_E E_0 = \frac{\partial E}{\partial V} \sigma_V V_0 = N(d_1)\sigma_V V_0$$

This equation together with the option pricing relationship enables $V_0$ and $\sigma_V$ to be determined from $E_0$ and $\sigma_E$. This PD ($N(-d_2)$) in theory is a *risk-neutral probability* bc it is calculated from an option pricing model. PD are higher in risk-neutral words than in real world because the expected growth rate is the *$r_f$ rate*, which is lower than the growth rate of company's assets in the real life (prob of asset value dropping below the value of D is higher in risk-n. world, bc higher growth rate means asset values should have a higher drop than with lower growth rate like rf). The PD increases with the amount of debt $D_T$, with the volatility $\sigma_V$ of the company's assets and decreases with the initial value of the company's asset $V_0$.

*Disadvantages:*

- does not take into account extreme or rare events since it assumes normal distribution
- default occurs only at maturity, but in real life it can occurs even before (Black-Cox First-passage model)
- default corresponds to the liquidation of the company from the market (reorganization is an option in certain legislations)

**Credit Risk Models**

*Default correlation* is used to describe the tendency for two companies to default at about the same time (e.g. companies in the same industry or the same geographic region tend to be affected similarly by external events). Default correlation means that *credit risk cannot be completely diversified away*, and it is the major reason why risk-neutral default probabilities are greater than real-world default probabilities.

There are three main quantitative approaches to analysing credit risk. In the ***structural approach***, we make explicit assumptions about the dynamics of a firm's assets, its capital structure, and its debt and shareholders. A *firm defaults if its assets are insufficient* according to some measure. In this situation a *corporate liability can be characterized as an option* on the firm's assets. The ***reduced form approach*** is silent about why a firm defaults. Instead, the dynamics of default are exogenously given through a *default rate*, or intensity. In this approach, *prices of credit sensitive securities can be calculated as if they were default free* using an interest rate that is the risk-free rate adjusted by the intensity. The ***incomplete information approach*** combines the best features of the two approaches.

**Structural models (Merton, Vasicek (KMV), Black and Cox)**

The basis of the structural approach is that *corporate liabilities are contingent claims on the assets of the firm*. The market value of the firm is the fundamental source of uncertainty driving credit risk. These models explain corporate defaults with the asset value of the firm falling below a *threshold*.

They are called structural models as default is based on the company's balance sheet structure. Default correlation between company A and B is introduced by assuming that the stochastic process followed by the assets of company A is correlated with the stochastic process followed by the assets of company B.

*Advantage*:

- They allow us to *use option pricing formulas* such as Black Scholes option pricing methods to evaluate default probability and recovery rate.

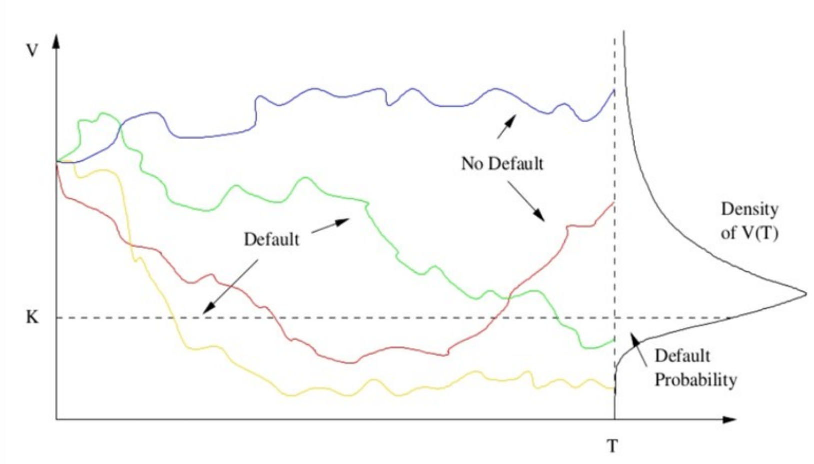
- The Merton model is best suited for companies whose *shares are publicly traded*.

## Merton Model

The main idea is that the *equity of the firm is an option on company's assets.* The firm is financed by both equity and debt, and debtholders have priority.

Assumptions:

- company's assets are tradeable in frictionless markets that are arbitrage-free
- risk-free interest rate is constant over time
- value of the firm's assets follows Geometric Brownian Motion (has lognormal distribution)
- the firm can neither repurchase nor issue new senior debt




The model assumes that the share of a leveraged company equals to a *call option*, where the underlying is the company's asset value. The strike price is the face value (notional) of the loans/debt (D). Maturity is the maturity of the loan (T). Simultaneously, owning the firm's debt is equivalent to *owning a riskless bond* that pays D at time T with certainty and *selling a European put option* on the assets of the company with a strike price of D at expiry T.

If the asset value $V_T$ exceeds or equals the face value K of the bonds, the bond holders will receive their promised payment K, and the shareholders will get the remaining $V_T$ - K. However, if the value of assets $V_T$ is less than K, the ownership of the firm will be transferred to the bondholders, who lose the amount K - $V_T$. Equity is worthless because of limited liability. Thus, the payoff at maturity is:

| | Assets | Bonds | Equity |
|---|---|---|---|
| No Default | $V_T \geq K$ | $K$ | $V_T - K$ |
| Default | $V_T < K$ | $V_T$ | 0 |

The value of equity and debt at maturity are:

$$S_T = \max[V_T - D, 0] \text{ and } B_T = \min[D, V_T]$$

The formula for the probability of default can be calculated as:

$$p(T) = P[V_T < K] = P[\sigma W_T < \log L - mT] = \Phi\left(\frac{\log L - mT}{\sigma\sqrt{T}}\right)$$

Moody's KMV model transforms the default probability produced by Merton's model into a *real-world default probability*, which is called Expected Default Frequency (EDF).

### KMV Model

KMV, by Moody's Analytics, builds upon the Merton model and modifies it to calculate the probability of default. They use a concept called *Distance to Default* to assess the risk. This distance is the difference between the value of the company's assets and the point at which default would occur, all relative to the volatility of the asset's returns. The market information contained in the firm's stock price and balance sheet are translated into an implied risk of default. It uses market observable data ($E_0$ and $\sigma_E$) to calculate the $V_0$ and $\sigma_V$ (from implied asset volatility) to get a distance to default.

Steps:

- Estimation of the market value and volatility of the firm's assets.
- Calculation of the distance to default, an index measure of default risk.
- Scaling of the distance to default to actual probabilities of default using a default database.

Merton's Model:

$$E_0 = V_0 * N(d_1) - D * e^{-rT} * N(d_2)$$

where

1) $d_1 = \dfrac{\ln\left(\frac{V_0}{D}\right) + \left(r + \frac{\sigma^2}{2}\right) * T}{\sigma\sqrt{T}}$ and $d_2 = d_1 - \sigma\sqrt{T}$

2) $\sigma_E E_0 = \sigma_V V_0 \Delta, \textit{where } \Delta = N(d1)$

**DD (Distance-to-default)** is the *number of SD the asset price must change for default* to be triggered T years in the future. It is $d_2$. As the DD declines, the company becomes more likely to default.

$$\text{DD} = \frac{\ln\left(\frac{V_0}{D}\right) + \left(r - \frac{\sigma_V^2}{2}\right) * T}{\sigma_V\sqrt{T}}$$

Simplified to $DD=\dfrac{V_t-D}{\sigma_V V_t}$

**Expected default frequency (EDF)** is a forward-looking measure of actual probability of default. EDF is firm specific. It is the *expected probability that a given counterparty default within one year*.

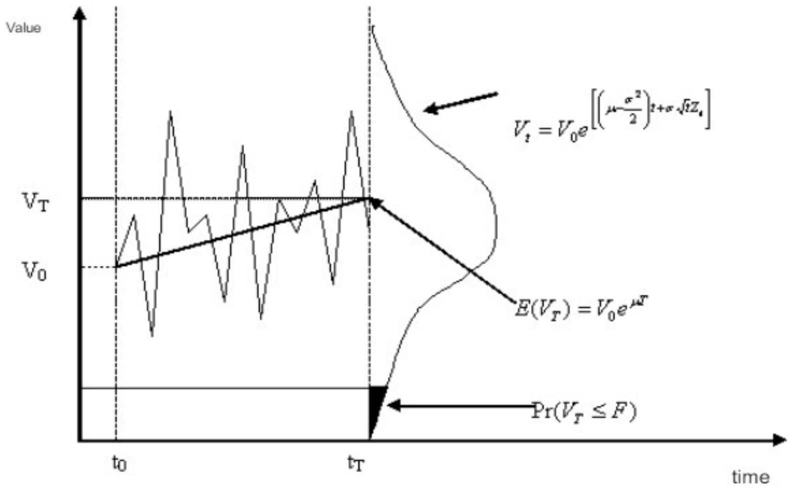
$$\text{EDF} = \Theta\left(-DD\right) = \Theta\left(-d_2\right)$$

In a risk-free world: EDF = $N(-d_2) = 1 - N(d_2)$

Based on historical information on a large sample of firms, for each time horizon, one can estimate the proportion of firms of a given default distance (say, df = 4.0) which actually defaulted after one year.

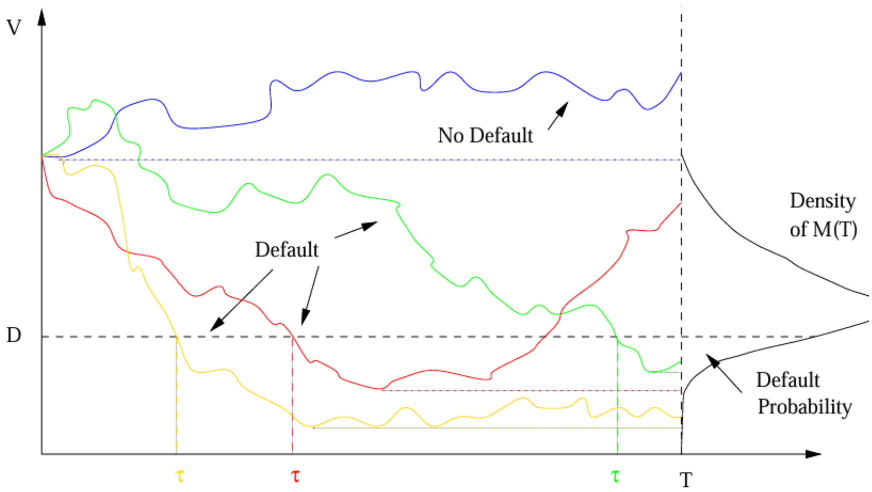
*Advantages:*

- Changes in EDF tend to anticipate at least one year earlier than the downgrading of the issuer by rating agencies. So the model allows default within one year.
- Gives more up-to-date information than periodically updated credit ratings.




## Black and Cox Model (First-passage approach)

The simplest first passage model takes a firm with asset value following GBM and outstanding debt with face value K at maturity T. In the Merton model, firm value can move to almost nothing without triggering default. Here, instead of admitting only the possibility of default at maturity time T, Black and Cox (1976) argues that *default occurs at the first time that the firm's asset value drops below the face value of the debt*. This can be explained by the right of bondholders to exercise a "safety covenant" that allows them to liquidate the firm if at any time its value drops below the specified threshold K(t).




If the firm value never falls below the barrier D, as D >= K, over the term of the bond ($M_T$ > D), then bond investors receive the face value K < $V_0$ and the equity holders receive the remaining $V_T$ - K. However, if the firm value falls below the barrier at some point during the bonds term ($V_T$ <= D), then the firm defaults. In this case the firm stops operating, bond investors take over its assets D and equity investors receive nothing. *Bond investors are fully protected*: they receive at least the face value K upon default and the bond is not subject to default risk anymore. However, if D < K then bond holders are exposed to default risk.

## Leland Model (Endogenous point of default)

In this model the firm owners/managers decide when to go into bankruptcy. The model incorporates the *dynamic trade-off theory*: tax benefits and cost of financial distress are also considered.

Firm value = asset value + PV(tax shield) - PV(financial distresses)

- $v(V, V_B, C) = V + TB(V, V_B, C) - BC(V, V_B, C)$
- $v(V, V_B, C) = E(V, V_B, C) + D(V, V_B, C)$

The probability of default is:

- $p_B(V) = \left(\frac{V}{V_B}\right)^{-\gamma}$
- where $\gamma = \frac{r}{\frac{1}{2}\sigma^2}$

The owners choose the optimal threshold ($V_B$) so the value of the equity is the highest.

- $V_B^*(C) = C\frac{\gamma(1-\tau)}{r(1+\gamma)}$

## Reduced form models

Here we assume that default occurs randomly, without warning at an exogenous default rate, or intensity. Instead of asking why the firm defaults (no assumption on the causes of default of a company), the intensity model is calibrated from market prices.

Assume that the hazard rates for different companies follow stochastic processes and are *correlated with macroeconomic variables and are no longer tied to the firm's assets*. When the hazard rate for company A is high there is a tendency for the hazard rate for company B to be high. This induces a *default correlation* between the two companies.

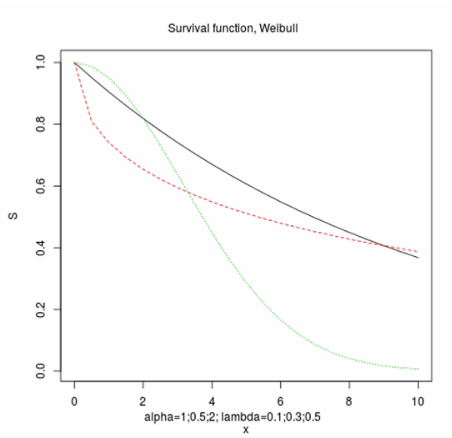
### Survival models

The ***survivor function***, denoted as S(t), represents the probability that the default time τ is greater than a specific time *t*. In other words, it defines the probability that and event (in this case a default) has not occurred before time t:

$$S(t) = Prob(\tau > t)$$

The survivor function is typically a decreasing function over time, starting from 1 at t=0 and approaching 0 as t increases.

| | 1 | 2 | 3 |
|---|---|---|---|
| 1 | 1.00 | 1.00 | 1.00 |
| 2 | 0.95 | 0.81 | 0.99 |
| 3 | 0.91 | 0.74 | 0.95 |
| 4 | 0.86 | 0.69 | 0.89 |
| 5 | 0.82 | 0.65 | 0.82 |
| 6 | 0.78 | 0.62 | 0.73 |
| 7 | 0.74 | 0.59 | 0.64 |
| 8 | 0.70 | 0.57 | 0.54 |
| 9 | 0.67 | 0.55 | 0.45 |
| 10 | 0.64 | 0.53 | 0.36 |

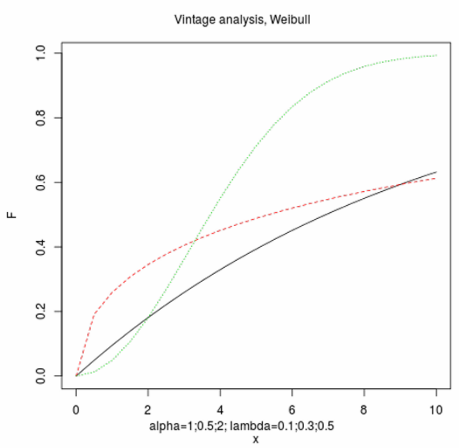



The ***cumulative probability distribution function*** (CDF) on default, denoted as F(t), represents the probability that the default occurs on or before a specific time t. It is the complement of the survivor function:

$$F(t) = Prob(\tau \leq t) = 1 - S(t)$$

The CDF starts at 0 and increases to 1 as time progresses, reflecting the increasing probability of default over time.

| | 1 | 2 | 3 |
|---|---|---|---|
| 1 | 0.00 | 0.00 | 0.00 |
| 2 | 0.05 | 0.19 | 0.01 |
| 3 | 0.10 | 0.26 | 0.05 |
| 4 | 0.14 | 0.31 | 0.11 |
| 5 | 0.18 | 0.35 | 0.18 |
| 6 | 0.22 | 0.38 | 0.27 |
| 7 | 0.26 | 0.41 | 0.36 |
| 8 | 0.29 | 0.43 | 0.46 |
| 9 | 0.33 | 0.45 | 0.55 |
| 10 | 0.36 | 0.47 | 0.64 |

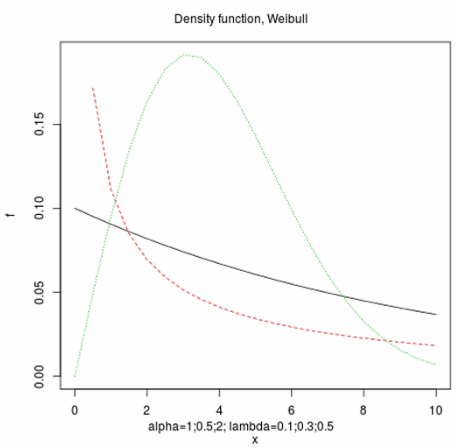



The ***probability distribution function*** (PDF) of the default time τ, denoted as f(t), describes the likelihood of default occurring at a specific time t. It is the derivative of the CDF:

$$f(t) = \frac{dF}{dt}(t)$$

The PDF provides information about when the default is most likely to occur.

| | 1 | 2 | 3 |
|---|---|---|---|
| 1 | 0.10 | Inf | 0.00 |
| 2 | 0.10 | 0.17 | 0.05 |
| 3 | 0.09 | 0.11 | 0.10 |
| 4 | 0.09 | 0.09 | 0.13 |
| 5 | 0.08 | 0.07 | 0.16 |
| 6 | 0.08 | 0.06 | 0.18 |
| 7 | 0.07 | 0.05 | 0.19 |
| 8 | 0.07 | 0.05 | 0.19 |
| 9 | 0.07 | 0.04 | 0.18 |
| 10 | 0.06 | 0.04 | 0.16 |




The ***hazard rate***, also known as the intensity or default intensity, is a crucial concept in intensity models. It is denoted as lambda and represents the instantaneous rate of default at time t, given that the entity has survived up to that point. Mathematically, it is defined as:

$$\lambda(t) = \frac{f(t)}{S(t)}$$

A ***counting process*** is a stochastic process that represents the cumulative number of events (defaults) that have occurred up to a given time t. It is denoted by N(t) and starts at 0 at time t = 0 and increases by 1 each time a default event occurs.

Properties:

- the process in non-decreasing over time, since the cum. number of default events cannot decrease
- its right-continuous, meaning there are no sudden jumps in its value except at times of defaults

Relationship with Default Time:

- The default time tau is a random variable representing the time when a default occurs. The counting process N(t) is linked to the default time tau such that it takes the value 0 when t < tau, and 1 if t > tau

Relationship with Hazard Rate:

- Hazard rate lambda is the instantaneous rate of default at time t, given survival up to that time.
- It provides  measure of how quickly defaults are like to occur at a given time
- The expected change in the counting process over a small time interval dt can be approximately lambda(t) * dt

Overall, counting process is connected to hazard rate and is fundamental for estimating the risk of default within a given time frame.

N(t) is a Poisson process with homogenous lambda parameter.

Advantages:

- The use of observed data means that historical estimation procedures can be employed to determine credit risk measures.
- Credit risk varies with the business cycle.
- There is no need to specify the firm's balance sheet.

Disadvantages:

- The range of default correlations that can be achieved is limited. Even when there is a perfect correlation between the hazard rates of the two companies, the probability that they will default during the same period is very low.
- Past market conditions may not reflect the future, and estimates derived from observed data may be inappropriate
- Reduced form models do not provide an economic explanation of why default occurs.
- Reduced form models treat default as a random event, which is not the case in reality.

## Credit VaR

Credit risk VaR is calculated to determine both regulatory capital and economic capital (own estimate of capital for the risk taking). CVaR is the *credit risk loss over a certain period of time that will not be exceeded with a certain CI*. The time horizon for CVaR is often longer than that for Market risk VaR (1 year vs 1 day). Historical simulation is used to calculate MVaR. An important aspect of CVaR is *credit correlation*. As credit correlation increases (mostly in stressed economic conditions), companies will become more likely to default as they are similarly affected by economic factors.

### Credit VaR methodologies

**1)** Structural/„Mark-to-Market" models

- There is an *economic model* behind default (Merton)
- Simpler normality assumptions: Vasicek ((IRB) model)
- Full rating migration matrix estimate: CreditMetrics

### CreditMetrics and Rating Transitions Matrices

The *CreditMetrics approach,* proposed by JPMorgan, is based on transition matrices. These are matrices showing the probability of a company migrating from one rating category to another during a certain period of time. They are based on historical data produced by rating agencies. CreditMetrics involves carrying out a Monte Carlo simulation to estimate a probability distribution of credit losses. This distribution can be used to calculate credit VaR.
Vasicek's model and CreditRisk+ estimate the probability distribution of losses arising from defaults. But the impact of *downgrades* in not considered. Here the default threshold is defined using credit ratings not liabilities as in the Merton's model. Thus, the model considers losses from both defaults and downgrades. Its disadvantage is that it is computationally time consuming.

### Vasicek's Model

The model is used in the Basel II's IRB approach. It is used for calculating VaR for a portfolio of loans. Each loan has a PD by time T, and the *worst case default rate* (WCDR) that will not be exceeded at the X confidence level is:

$$\mathrm{WCDR}(T, X) = N\left(\frac{N^{-1}(\mathrm{PD}) + \sqrt{\rho}N^{-1}(X)}{\sqrt{1-\rho}}\right)$$

For an individual loan, the VaR is:

$$\mathrm{WCDR}(T, X) \times \mathrm{EAD} \times \mathrm{LGD}$$

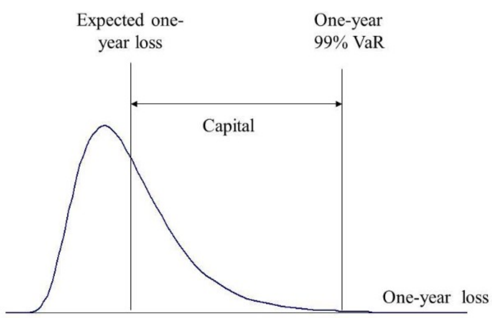
For a portfolio of loans, the **VaR** is:

$$\sum_{i=1}^{n} \mathrm{WCDR}_i(T, X) \times \mathrm{EAD}_i \times \mathrm{LGD}_i$$

The required capital is:

$$\sum_i EAD_i * LGD_i * \left(WCDR_i - PD_i\right)$$

*Regulatory capital* for banks is calculated with **T = 1 year and X = 99.9%**. It is the difference between the EL and VaR.



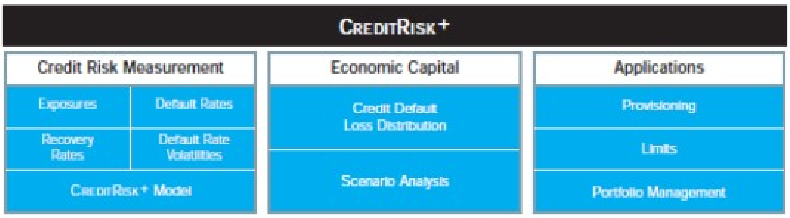

Vasicek's model has the *disadvantage* that it incorporates very little tail correlation. Replacing the normal distribution can remedy this.

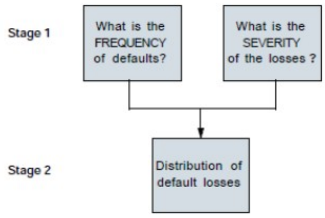
**2)** „Default mode"/"Actuary" models

- **The reason of default is not observable**
- Recursive summation of elementary distributions
- Requires PD standard deviation in addition to PDs: CreditRisk+

**CreditRisk+**

It is a methodology to calculate CVaR developed by Credit Suisse involving analytic approximations. Uses similar procedures to those used in the insurance industry to calculate default losses from assumptions about the probabilities of default of individual companies.




The model allows for the modelling of complex portfolios of instruments. Default probability is conditionally Poisson distributed. Introduces correlation and dependence among defaults, variable default rates and economic trends and cycles.

- Default distribution:
  - The model assumes that default events are independent. In practice we do not know the future default rate, but we can assume what the *expected number of defaults* will be (for 1 year). If default rate is assumed to be known (no PD volatility) we can use a Poisson distribution to determine the probability of *m* defaults. If we assume the default rate in unknown, the expected number of defaults will have a *gamma distribution* (there is PD volatility), and the probability of *m* defaults will have a negative binomial distribution. As the uncertainty about default rate increases, the correlation increases. With default correlation, the loss probability distribution is positively skewed.
  - Using one of these distributions (Poisson, gamma) we can determine the portfolio loss distribution by aggregating losses across all obligors. Then we can calculate the CVaR w/ the desired CI.

## Stress Testing

Credit risk stress testing is a financial risk management technique used to assess the potential impact of adverse economic conditions on a financial institution's credit portfolio. It involves simulating extreme economic scenarios and analyzing how these scenarios would affect the creditworthiness of borrowers and, consequently, the institution's overall financial health.

Key Goals of Credit Risk Stress Testing:

- *Identify vulnerabilities*: Pinpoint weaknesses in the credit portfolio that could lead to significant losses.
- *Assess capital adequacy*: Determine if the institution holds sufficient capital to withstand extreme economic conditions.
- *Regulatory compliance*: Meet regulatory requirements for stress testing and capital planning.

By understanding the potential impact of adverse events, financial institutions can make informed decisions to strengthen their risk management practices and protect their financial stability.

*Climate risk* has emerged as a critical factor in credit risk stress testing due to its potential to significantly impact the economy, financial markets, and individual borrowers. There are two primary ways climate risk intersects with credit risk stress testing:

1. Physical Climate Risks

- Extreme weather events: Hurricanes, floods, wildfires, and other extreme weather events can cause substantial property damage, business interruptions, and economic losses, increasing the likelihood of loan defaults.
- Sea level rise: Coastal properties and businesses are at risk from rising sea levels, leading to increased insurance costs, property devaluation, and potential financial distress.

2. Transition Risks

- Policy changes: Governments implementing stricter environmental regulations or carbon taxes can lead to significant costs for businesses, potentially impacting their creditworthiness.
- Market shifts: Changes in consumer preferences towards sustainable products and services can impact the profitability of companies that fail to adapt.

**Counterparty Credit Risk** *(Risk Management course – Lecture 10 slides)*

Counterparty credit risk (CCR) is the *risk that the counterparty to a financial transaction could default before the final settlement of the transaction's cash flows / before the expiration of the contract*. An economic loss would occur if the transactions or portfolio of transactions with the counterparty has a *net positive economic value* at the time of default. CCR creates a *bilateral* *risk of loss*: the market value of the transaction can be positive or negative to either counterparty to the transaction. The market value is uncertain and can vary over time with the movement of underlying market factors.

In the Over-The-Counter (OTC) market, transactions are subject to counterparty risk due to the absence of a *central clearinghouse*, whereas in exchange-traded markets, transactions are cleared through a centralized clearinghouse, reducing counterparty risk.
In OTC markets, transactions can occur in two ways: through central counterparties (CCPs) or bilaterally. Under the new regulations, standardized derivatives must be transacted through CCPs, and bilateral agreements are only allowed with "non-systemically important" clients. Counterparty risk arises only from the latter type of transactions. In case of CCPs, there is a requirement of *initial margin* and *variation margin* from members. Assessing the magnitude of this risk is much more challenging than in the case of credit transactions, as exposure in derivatives changes daily relative to our counterparties.

**Methods to Reduce Counterparty Risk:**

- **Netting**: Agreement between two parties to net out transactions: all transactions are considered to be a single transaction in the event of default.
- **Margin agreements**: Agreement requiring both parties to post a certain amount as collateral (haircut); if exposure not covered by collateral exceeds a threshold.
- **Downgrade clauses:** If the counterparty's rating falls below a certain category, the bank has the option to exit the contract at market value. (Example: Enron, 2001)
- **Central Counterparty (CCP):** Performs broader and more efficient netting, requires margin from clients.

**CVA and DVA**

**Credit Value Adjustment (CVA)** is the bank's estimate of the PV of the expected loss to the bank of a counterparty default. It is the amount by which the bank should reduce the value of transactions because of counterparty default risk.

$$\text{CVA} = \sum_{i=1}^{N} q_i v_i$$

where $q_i$ is the default probability and $v_i$ is the PV of the avg net exposure at $i$th default time.

**Debit (or debt) valuation adjustment (DVA)** is the PV of the expected gain to the bank from the possibility that it might itself default. It is the cost to the counterparty of a default by the bank.

$$\text{DVA} = \sum_{i=1}^{N} q_i^* v_i^*$$

Taking both CVA and DVA into account, the value of the portfolio to the bank against the counterparty is:

$$f_{\text{nd}} - \text{CVA} + \text{DVA}$$

As the bank's creditworthiness declines, $q^*_i$ (PD of the bank) increases and DVA increases. This makes the derivatives portfolio more valuable to the bank. As the bank's creditworthiness improves, $q^*_i$ decreases and DVA decreases so that the portfolio is less valuable.

CVA and DVA must be calculated on the whole portfolio of derivatives that a bank has with a counterparty.

**Wrong Way Risk** occurs when $q_i$ is positively correlated with $v_i$. Occurs for ex. when CP is selling credit protection.

**Right Way Risk** occurs when $q_i$ is negatively correlated with $v_i$. Occurs for ex. when CP is buying credit protection.

## Regulation

The purpose of bank regulation is to ensure that banks keep enough capital for the risks they take. The fundamental question for banks is therefore the measurement of *capital adequacy*: do they have enough capital to cover future losses? The problem is with systematic risk, i.e. the risk that the failure by a large bank will lead to failure by other large banks and finally the financial system.

**1) Basel I** (1988)

It set international risk-based standards for capital adequacy.

- ***Capital Adequacy Ratio***
    - ratio of a bank's capital to its *risk-weighted assets (RWAs)* - banks were required to assign a risk weight to each asset class, reflecting the asset's credit risk
    - capital consisted of *Tier 1* (common stock, RE) and *Tier 2* (subordinated debt) capital

- ***minimum capital requirement***
    - at least 8% of risk-weighted assets (RWA), at least half of it coming from Tier 1 capital

Critics: not risk-sensitive - regulatory capital has little to do with risk

- Ad hoc weights - not dependent on the bank's risk management and portfolio quality
- Deals only with credit risk
- The question "what is capital" is more important than "how much should capital be?

**2) Basel II** (2004, implemented in 2007)

It is a more risk-sensitive approach to credit risk management and capital adequacy requirements. It has three pillars:

- *New minimum capital requirements for credit and operational risk*
    - Banks could choose between risk weights based on either external credit ratings (standardized approach) or on bank's own internal credit ratings (IRB approach)
    - Separate capital charge for operational risk

Credit risk measurement methods (Pillar I): Standard Approach, Foundation IRB Approach, Advance IRB Approach.

- Standard method (STA) – regulatory RWs
    - simplest methodology, continuation of Basel I, being based on the rating of debtors by an external rating agency and the ad hoc (arbitrary) Risk Weight (RW) assigned to it
    - risk weights specified by the regulator
- Internal Rating Based (IRB) method
    - based on internal models/rating
    - Credit VaR formula is subject to supervisory authorization
    - Capital is based on a 1-year VaR at 99.9% CI. Since banks usually cover expected losses (higher interest rate for a worse client, etc.), the required capital is the calculated VaR – EL. It is the same as:

$$\sum_i \mathrm{EAD}_i \times \mathrm{LGD}_i \times (\mathrm{WCDR}_i - \mathrm{PD}_i)$$

    - Key parameters: PD, LGD, EAD, M (maturity adjustment factor)
      <u>Foundation IRB</u>: PD estimated by the bank, the rest set by regulators

Advanced IRB: all of them estimated by the company

- ***rho** is always set by regulators*

3) **Basel III** (2010)

- **Minimum Leverage Ratio**
  - Ratio of **Tier 1 capital to Total Exposure** must be **greater than 3%.**
  - Exposure includes all items on-balance sheet, derivatives exposures (calculated as in Basel I), securities financing exposures, and some off-balance sheet items
- **Credit Value Adjustment (CVA)**
  - Used in *bilateral clearings* and calculated for each of its derivative counterparties
  - *The expected loss for the bank from the default of a counterparty*
  - Reported profit is reduced by CVA
  - It comes from a parallel shift in the term structure of counterparty's credit spreads; CVA arising from the changing credit spreads is required to be a component of *market risk capital*
  - *Wrong-way risk*: positive relationship is assumed between PD of a counterparty and exposure to this counterparty; ex. when a counterparty is using CDS to sell protection to a dealer
  - *Right-way-risk*: negative relationship is assumed between PD of a counterparty and exposure to this counterparty; ex. when a counterparty is using CDS to buy protection from a dealer
- **Debit Value Adjustment**
  - Used in *bilateral clearings*
  - *Expected loss to the counterparty from the default of the bank*
  - The counterparty's CVA